Day 24 Problem Set

Geometry Summer Credit | Coordinate Geometry Synthesis

Name:	Date:	Period:

Directions

Show coordinate evidence: slope, distance, midpoint, or equation features. Label conclusions clearly.

Part A: Lines, Distances, and Midpoints

1.	Find the slope through (1, 4) and (7, 16).

2.	Find the slope through (-2, 9) and (4, -3).

3.	Are lines with slopes 3/5 and 3/5 parallel, perpendicular, or neither?

4.	Are lines with slopes 4/7 and -7/4 parallel, perpendicular, or neither?

5.	Find the distance between (-5, 2) and (7, 7).

6.	Find the midpoint of (8, -6) and (-4, 10).

7.	A triangle has vertices A(0, 0), B(6, 0), and C(6, 8). Find its perimeter.

Part B: Circles and Coordinate Classifications

8.	Write the equation of a circle with center (-2, 5) and radius 9.

9.	A circle has equation $(x + 4)^2 + (y - 1)^2 = 36$. Name the center and radius.

10.	A circle has center (1, -3) and passes through (7, 5). Find its radius.

11.	A circle has diameter endpoints A(-4, 6) and B(8, -2). Find the center.

12.	For the circle in item 11, find the radius.

13.	Use slopes to decide whether A(1, 1), B(7, 1), C(7, 5), D(1, 5) form a rectangle.

14.	Use side lengths to decide whether the rectangle in item 13 is a square.

Day 24 Problem Set

Continued | Algebra 2 Bridge

15.	Describe the transformation from $y = x^2$ to $y = (x - 6)^2$.

16.	Describe the transformation from $y = x^2$ to $y = x^2 - 4$.

17.	Describe the transformation from $x^2 + y^2 = 16$ to $(x + 3)^2 + (y - 2)^2 = 16$.

18.	Spiral review: Two similar solids have scale factor 2. The smaller surface area is 30. Find the larger surface area.

19.	Spiral review: A cylinder has radius 5 and height 8. Find its volume in terms of pi.

20.	Challenge: A circle has equation $x^2 + y^2 - 6x + 8y = 0$. Complete the square to find its center and radius.